

Kenneth Li

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Professional Summary

- I am a data-driven Assistant Engineering Geologist with a Master's in Geomatics and with five years of experience in geospatial data analysis, automation, and visualization at Arup.
- My experience covers a wide range of tasks in geotechnical projects including aerial photographs interpretation, slope safety upgrading works, geological modelling and site supervision of GI works at locations with complex geology. I have been responsible for desk studies, engineering geological assessments with the use of LiDAR data, Geographical Information System (GIS), SQL, gINT, Oasys Slope, LS-Dyna, Leapfrog Works for geological modelling and PySpark. I also act as the Chairperson in the Geological Society of London, Hong Kong Regional Group (Young Fellows Committee), to help organize various activities, field trips, technical talks and engaging with professionals.

Professional Experience

JULY 2021 – PRESENT

Assistant Geologist | Arup

- Designed and implemented automated data extraction pipelines using Python, improving geospatial data processing efficiency by 95% for groundwater monitoring, enabling real-time reporting for stakeholder decision-making.
- Developed interactive GIS-based dashboards using ArcGIS and borehole data shapefiles, integrating georeferenced aerial imagery to visualize terrain data, enhancing site supervision and stakeholder communication for geotechnical projects.
- Conducted advanced spatial analysis using 3D LiDAR data (0.5m resolution) to model natural terrain landslide risks, generating actionable insights for risk assessment and mitigation strategies.
- Utilized handheld laser scanners to acquire high-resolution 3D data for slope stabilization, contributing to Building Information Modeling (BIM) and supporting infrastructure project planning.
- Collaborated with multidisciplinary teams to integrate geospatial analytics into project workflows, aligning data solutions with operational needs and driving adoption of analytical tools.
- Supervised site works for Public Housing Development project, overseeing the completion of 262 drillholes during my supervision period and coordinating with the client, design office and 4 contractors. I also used cleaned AGS data provided by the contractors with Leapfrog Works to monitor the abrupt changes in engineering rockhead compared with the surrounding ground investigation data to find out the related geotechnical risks

JUNE 2020 – AUGUST 2020

Engineering Geology Intern | Geotechnical Engineering Office | Hong Kong

- Developed a Python-based digital mapping system to streamline geospatial data processing, supporting open mapping initiatives and improving data accessibility for community-driven projects.
- Analyzed limitations of LiDAR technologies, providing insights to optimize data collection strategies for high-accuracy geospatial applications.

Education

SEPTEMBER 2023 – MAY 2025

Master of Science – Geomatics | Hong Kong Polytechnic University | Hong Kong

- 3.80 CGPA (Distinction)
- Dissertation: “Further Enhancement in Rock Outcrop Recognition in Hong Kong with Deep Learning Approach Using Very High-Resolution Aerial Imagery”
- Built a deep learning model (88% accuracy) using Python, TensorFlow, and LabelMe to analyze geospatial imagery downloaded via API, supporting data-driven risk assessments for geotechnical applications.
- Experience in using PySpark for geospatial clustering to analyze the possibility of shared cab in NYC, the dataset covered more than 173 million individual taxi trips taken in New York City during 2013

SEPTEMBER 2017 – DECEMBER 2021

Bachelor of Science (Major in Geology, Minor in Computer Science) | The University of Hong Kong | Hong Kong

- Award: Geological Society of Hong Kong Best Student Prize in Earth Sciences
- Dissertation: “Detection of Change in Geometry and Landform Features of Landslides Using 4-Dimensional Remote Sensing Data”

Leadership Activities

AUGUST 2024 - PRESENT

Chairperson of Young Fellows Group - Geological Society of London (Hong Kong Regional Group)

- Organized seminars, competitions, and an annual dinner to foster collaboration within the geotechnical community, engaging 100+ professionals.
- Led a field trip for 70 participants to explore Hong Kong’s geological features, promoting knowledge-sharing and open data initiatives.

Technical Skills

- Geospatial Tools: ArcGIS, QGIS, Google Maps API, PySpark, 3D LiDAR data processing
- Programming: Python, Java (app development), automation scripting in Esri ArcGIS Pro/ArcMap
- Data Analysis: Deep learning (CNNs for geospatial imagery), geospatial data modeling
- Other: Building Information Modeling (BIM), open data frameworks, Project Coordination with different parties